Classic Compute Governance

> **Document Owner:** Platform Engineering | **Status:** Approved | **Last Updated:** February 2026

Overview

Serverless is mandatory first choice (see SC-01). This document governs **classic compute** for remaining use cases: GPU, 24/7 streaming, and workloads requiring specific Spark configurations. It covers access modes, cluster policies, Photon, and workload-based sizing.

Golden Rules

ID	Rule	Severity	WAF Pillar
CP-01	All classic clusters must use approved policies	Critical	Cost
CP-02	No unrestricted cluster creation	Critical	Cost
CP-03	Instance types limited by policy	Required	Cost
CP-04	Auto-termination required	Required	Cost
CP-05	Jobs compute for scheduled, All-Purpose for exploration	Critical	Cost
CP-06	Dependencies consistent between All-Purpose and Jobs	Required	Cost
CP-07	Use standard access mode for most workloads	Required	Security
CP-08	Enable Photon for complex transformations	Required	Performance
CP-09	Right-size clusters based on workload type	Required	Cost

> **Note:** Serverless is mandatory first choice — see SC-01. This document covers the classic compute exception path only.

CP-01: All Classic Clusters Must Use Approved Policies

Rule

Every classic cluster (Jobs or All-Purpose) must be created under an approved cluster policy. No unrestricted cluster creation is permitted.

Why It Matters

- Policies enforce cost controls, instance limits, and auto-termination
- Prevents runaway compute costs from oversized or forgotten clusters
- Ensures consistent tagging for cost attribution

Standard Policies

1. Jobs - Data Engineering (CP-03)

```
{
  "name": "Jobs - Data Engineering",
  "definition": {
    "spark_version": {
      "type": "allowlist",
      "values": ["14.3.x-scala2.12", "15.4.x-scala2.12"]
    },
    "node_type_id": {
      "type": "allowlist",
      "values": ["i3.xlarge", "i3.2xlarge"]
    },
    "num_workers": { "type": "range", "minValue": 1, "maxValue": 10 },
    "autotermination_minutes": { "type": "fixed", "value": 10 },
    "cluster_type": { "type": "fixed", "value": "job" },
    "custom_tags.team": { "type": "required" },
    "custom_tags.cost_center": { "type": "required" }
  },
  "libraries": [/* standard libraries */]
}
```

2. Jobs - ML Training GPU (CP-03)

```
{
  "name": "Jobs - ML Training GPU",
  "definition": {
    "spark_version": {
      "type": "allowlist",
      "values": ["14.3.x-gpu-ml-scala2.12"]
    },
    "node_type_id": {
      "type": "allowlist",
      "values": ["g4dn.xlarge", "g4dn.2xlarge"]
    },
    "num_workers": { "type": "range", "minValue": 1, "maxValue": 4 },
    "autotermination_minutes": { "type": "fixed", "value": 30 }
  },
  "libraries": [/* standard + ML libraries */]
}
```

3. All-Purpose - Development (CP-03)

```
{
  "name": "All-Purpose - Development",
  "definition": {
    "spark_version": {
      "type": "allowlist",
```

```
        "values": ["14.3.x-scala2.12", "15.4.x-scala2.12"]
      },
      "node_type_id": {
        "type": "allowlist",
        "values": ["i3.xlarge"]
      },
      "num_workers": { "type": "range", "minValue": 0, "maxValue": 2 },
      "autotermination_minutes": { "type": "fixed", "value": 15 },
      "cluster_type": { "type": "fixed", "value": "all-purpose" }
    },
    "libraries": [/* same as Jobs policy */],
    "max_clusters_per_user": 1
  }
```

Policy Enforcement

When policies are updated, existing clusters may be out of compliance:

1. View compliance: **Compute** > **Policies** > **Compliance** column
2. Fix compliance: Click **Fix all** to update all resources
3. Monitor: Check `system.compute.clusters` for policy assignments

CP-02: No Unrestricted Cluster Creation

Rule

Remove the `Unrestricted` cluster creation permission from all users and groups. All clusters must be created under an approved policy.

Why It Matters

- Unrestricted access allows users to create oversized, untagged clusters
- Bypasses all cost controls and instance type limits
- A single unrestricted cluster can consume an entire budget

Implementation

1. Navigate to **Admin Console** > **Workspace Settings** > **Cluster creation**
2. Set to **Policy-restricted** (remove Allow Unrestricted)
3. Assign appropriate policies to groups

CP-03: Instance Types Limited by Policy

Rule

Cluster policies must restrict instance types to an allowlist appropriate for the workload type.

Why It Matters

- Prevents selection of oversized or GPU instances for simple workloads
- Standardizes instance families for better reserved instance utilization
- Reduces the risk of accidental cost spikes

CP-04: Auto-Termination Required

Rule

All clusters must have auto-termination configured. No indefinitely running clusters except approved always-on streaming workloads.

Auto-Termination Settings

Environment	Setting
Development	15 minutes
Staging	30 minutes
Production (batch)	60 minutes
Production (streaming)	Disabled (0) — requires approval

Required Tags

All clusters must include required tags for cost attribution (see EA-05, TG-01):

```
{
  "custom_tags.team": { "type": "required" },
  "custom_tags.cost_center": { "type": "required" },
  "custom_tags.environment": {
    "type": "allowlist",
    "values": ["dev", "staging", "prod"]
  }
}
```

CP-05: Jobs Compute for Scheduled, All-Purpose for Exploration

Rule

Use Jobs compute for all scheduled/automated workloads. All-Purpose compute is restricted to interactive exploration and debugging only.

Comparison

Aspect	Jobs Compute	All-Purpose
Purpose	Scheduled/automated	Interactive exploration

Aspect	Jobs Compute	All-Purpose
Lifecycle	Created per run	Long-running
Use For	ETL, ML training, reports	Debugging, prototyping
Not For	Interactive work	Production workloads

Anti-Patterns

- Running production ETL on All-Purpose clusters (higher cost, no auto-scaling)
- Using Jobs compute for interactive debugging (poor developer experience)

CP-06: Dependency Consistency Between All-Purpose and Jobs

Rule

Code developed in All-Purpose must work identically in Jobs compute. Both environments must use the same library versions, enforced via cluster policies.

Why It Matters

- "Works on my cluster" failures are the #1 cause of deployment issues
- Library version mismatches cause silent data quality bugs
- Policy-enforced libraries ensure a single source of truth

Policy-Based Libraries

```
{
  "libraries": [
    { "pypi": { "package": "pandas==2.0.3" } },
    { "pypi": { "package": "pyarrow>=14.0.0" } },
    { "pypi": { "package": "scikit-learn==1.3.2" } },
    { "pypi": { "package": "mlflow>=2.17.0" } }
  ]
}
```

Effect: Users cannot install/uninstall libraries on compute using this policy.

CP-07: Use Standard Access Mode for Most Workloads

Rule

Use standard (shared) access mode for most classic compute workloads. Dedicated access mode is only for GPU, RDD APIs, R, or container services.

Why It Matters

- Multi-user with full user isolation — more cost-effective through resource sharing
- Enforces Unity Catalog governance automatically
- Reduces cluster sprawl (one shared cluster vs. many single-user clusters)

Decision Guide

```
Need GPU, RDD, R, or Container Service?
├── YES → Dedicated Access Mode (SINGLE_USER)
└── NO → Standard Access Mode (USER_ISOLATION)
```

Access Mode Comparison

Feature	Standard (Shared)	Dedicated (Single-User)
Multi-user	Yes	No
User isolation	Yes	N/A
Unity Catalog	Required	Required
RDD APIs	No	Yes
GPU support	Limited	Full
R language	No	Yes

Implementation

```yaml
# Standard access mode (default)
new_cluster:
  data_security_mode: USER_ISOLATION

# Dedicated (only when required)
new_cluster:
  data_security_mode: SINGLE_USER
```

CP-08: Enable Photon for Complex Transformations

Rule

Enable Photon for workloads with complex joins, large aggregations, or frequent disk access. Photon is built into serverless compute; for classic clusters, it must be explicitly enabled.

Why It Matters

- Significant performance improvement for complex SQL and DataFrame operations
- Native vectorized query execution
- Optimized for wide transformations

> **Note:** Adaptive Query Execution (AQE) is enabled by default on Databricks Runtime 14.3+. Combined with Photon, AQE delivers significant performance gains without manual tuning.

When Photon Helps Most

Workload	Benefit
Complex joins	High
Large aggregations	High
Large table scans	High
Simple ETL (<2 sec queries)	Low

Implementation

```
new_cluster:
  spark_version: "14.3.x-photon-scala2.12"
  runtime_engine: PHOTON

# Or for DLT
pipelines:
  gold:
    photon: true
```

CP-09: Right-Size Clusters Based on Workload Type

Rule

Size clusters based on workload characteristics. Use fewer larger workers for shuffle-heavy jobs; use single-node for interactive analysis.

Why It Matters

- Over-provisioning wastes cost; under-provisioning causes failures and timeouts
- Instance type selection impacts shuffle performance, memory pressure, and cost
- Right-sizing is the most impactful cost optimization for classic compute

Sizing Guide

Workload	Workers	Instance Size	Photon
Data Analysis	0 (single node)	Large	Optional
Basic ETL	2-4	Small/Medium	No
Complex ETL	2-4	Large	Yes
ML Training	0-2	Large	No

Workload	Workers	Instance Size	Photon
Streaming	4-8 (fixed)	Medium	Optional

Auto-Scaling Guidelines

Workload	Auto-Scaling
Interactive notebooks	Yes
Variable batch ETL	Yes
Streaming	No (fixed size)
ML Training	No (fixed size)

Validation Checklist

Serverless-First

- [ ] Serverless evaluated first for ALL workloads
- [ ] Classic compute only when serverless not supported
- [ ] Exception documented if using classic

Compute Type Separation

- [ ] Jobs compute for scheduled workloads
- [ ] All-Purpose only for interactive exploration
- [ ] No production on All-Purpose

Dependency Consistency

- [ ] Same library versions in Jobs and All-Purpose
- [ ] Libraries defined in policies
- [ ] Single source of truth (requirements.txt)

Policy Design

- [ ] Instance types appropriate for workload
- [ ] Auto-termination configured
- [ ] Required tags defined
- [ ] max_clusters_per_user set for All-Purpose

Access Mode & Configuration

- [ ] Standard access mode unless GPU/RDD/R/Container required (CP-07)
- [ ] Photon enabled for complex transformations (CP-08)
- [ ] Cluster size matches workload type (CP-09)
- [ ] Auto-scaling enabled for variable workloads, disabled for streaming/ML

Related Documents

- Serverless Compute
- Platform Overview

References

- Compute Policies
- Add Libraries to Policy
- Compute Configuration Recommendations
- Photon
- Access Modes